

Introductory Game Theory, Lecture 1

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Fall 2026

Game Theory: What Is It about?

- A strategic situation is a state of affairs in which the *outcome* depends on the *decisions of all parties* (individuals, organizations, etc.).
- This implies that each party's *payoff* (i.e., the subjective evaluation of the outcome) depends on other parties' decisions.
- So strategic situations are those that feature *payoff interdependence*.
- Such payoff interdependence typically implies *strategic interdependence*, namely, how each party will behave relies on how other parties will behave.

Strategic situations are abundant in our real world.

Examples of strategic situations

- Wars
- Bidding in auctions
- Cellphone pricing
- Penalty kick in football matches

Examples of non-strategic situations

- Deciding what to eat for dinner today
- Choosing your consumption bundle on competitive markets

A brief history about thoughts:

- The investigation of strategic situations can be traced back to ancient times. The most stark example is the military treaties *The Art of War* by Sun Tzu (5th century BC), which is the first systematic theory about one of the most important strategic situations—war.
- The modern idea of strategic situation originated from *games* (by its literal meaning) like nim, chess, and hex, which are obvious examples that feature payoff interdependence.
- People later realized that all strategic situations could be analyzed under a uniform framework. Thus, the term "game" has been extended to mean all mathematical models of strategic situations.

- Under the uniform framework, people have developed a modern theory which investigates *rational decision making* in mathematical models of strategic situations (i.e., games).
- Such a theory is thus called *game theory*.
- Uniform framework:
 - (i) The uniform representation of key factors of a strategic situation, which transforms the original situation into a game;
 - (ii) Exploring the implication of *rationality* in games, which generates a general principle of "reasonable outcomes", called *solution concepts*.

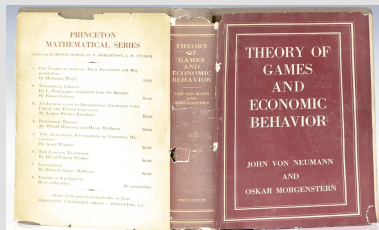
- Early development of game theory was not systematic and was scattered in the discussion of concrete cases.
 - Military books: *The Art of War*, *Sun Bin's Art of War*, *Thirty-six Stratagems*, etc.;
 - J. Waldegrave (1713) proposed a *minimax mixed strategy solution* to a two-person card game;
 - E. Zermelo (1913) proved that for games like chess there exists a strategy that guarantees that some player will never lose;
 - E. Borel (1938) proved a *minimax theorem* for a special two-person zero-sum game;
 - A. Cournot (1838) and J. Bertrand (1883) each proposed a model of *oligopolistic competition* and a solution.

A BRIEF HISTORY OF GAME THEORY (CONT.)

- The birth of modern game theory: *On the Theory of Games of Strategy* (1928) by J. von Neumann (1903-1957), and the publication of the book *Theory of Games and Economic Behavior* (1944) by J. von Neumann and O. Morgenstern (1902-1977).



Morgenstern and von Neumann



Theory of Games and Economic Behavior

- Laid down the foundations for modern game theory (unified framework, basic notions, and paradigm of analysis);
- Focusing on a particular class of games: *zero-sum games*.

A BRIEF HISTORY OF GAME THEORY (CONT.)

- A landmark in the development of game theory: *Equilibrium Points in n -person Games* (PNAS, 1950) by J. Nash (1928-2015).



John F. Nash

EQUILIBRIUM POINTS IN N -PERSON GAMES

By JOHN F. NASH, JR.*

PRINCETON UNIVERSITY

Communicated by S. Lefschetz, November 16, 1949

One may define a concept of an n -person game in which each player has a finite set of pure strategies and in which a definite set of payments to the n players corresponds to each n -tuple of pure strategies, one strategy being taken for each player. For mixed strategies, which are probability

Title of the paper *Equilibrium Points in n -person Games*

- Proposed the fundamental solution concept of *Nash equilibrium*, which is widely applicable to strategic analysis (not just zero-sum games);
- Laid down the principle of rationality in strategic situations;
- Most modern development in solution concept is based on Nash equilibrium.

Formal Ingredients of Strategic Situations

FOUR BASIC INGREDIENTS

- There are four key factors for describing a strategic situation:
 1. The set of involved parties, called *players* or sometimes *agents*
 2. The set of feasible *actions/choices* for each player
 3. The *payoffs* associated with each possible outcome
 4. The *information/knowledge* players have about the three factors above, others' decisions, and others' knowledge/information
- The four ingredients form the *structure* of a strategic situation.

SOME CLARIFICATION

- A player is not necessarily an individual, but a decision-making entity that is appropriately defined when setting up the model.
- A player's payoff from an outcome is his/her evaluation (satisfaction/desirability/utility) of the outcome. We often assume that every player evaluates an uncertain outcome by its expected payoff.
- A player's knowledge needs to specify not only what he/she knows about external parameters (like payoffs), but also what he/she knows about the other players' knowledge and beliefs about these parameters, as well as what he/she knows about the other players' knowledge of his/her own belief, and so on.
- Unless declared otherwise, we will assume that the set of players, players' feasible actions, and players' payoffs are *common knowledge*:
 - Every player knows it;
 - Every player knows that every player knows it;
 - Every player knows that every player knows that every player knows it;
 - ... (ad infinitum)

Consider the situation that you (A) and your friend (B) play rock-paper-scissors.

- Set of players: $\{A, B\}$;
- Feasible set of actions for each player: $a_A, a_B \in \{R, P, S\}$, a_i being the action for player i , $i \in \{A, B\}$;
- Payoffs: $u_A(a_A, a_B)$ and $u_B(a_A, a_B)$, where

$$u_A(R, S) = u_A(S, P) = u_A(P, R) = 1$$

$$u_A(R, R) = u_A(P, P) = u_A(S, S) = 0$$

$$u_A(S, R) = u_A(P, S) = u_A(R, P) = -1$$

and $u_B(a_A, a_B) = -u_A(a_A, a_B)$ for every (a_A, a_B) .

- The above factors are common knowledge.